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SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

● **Experiment No.: 5**

1. Title: Implement localStorage and sessionStorage to save user preferences like theme settings.

2. Software/Tools Required: VS Code, JavaScript,html.

3. Theory:

- 1) Web Storage allows websites to store data in the browser using key-value pairs.
- 2) localStorage stores data persistently and remains available even after closing the browser.
- 3) sessionStorage stores data only for the current browser session.
- 4) setItem() is used to save the selected theme and storage type.
- 5) getItem() retrieves previously saved theme and storage information.
- 6) removeItem() removes stored preferences.
- 7) The program provides Light, Dark, and Blue themes using CSS classes.
- 8) classList.remove() and classList.add() are used to change the theme dynamically.
- 9) getElementById() accesses the required HTML elements.
- 10) innerText displays the current theme, storage type, saved value, and messages.
- 11) loadPreference() automatically retrieves saved preferences when the page loads.
- 12) savePreference() stores the selected theme in either localStorage or sessionStorage.
- 13) removePreference() clears the saved preferences and resets the theme to Light.
- 14) The program demonstrates web storage and dynamic DOM manipulation.



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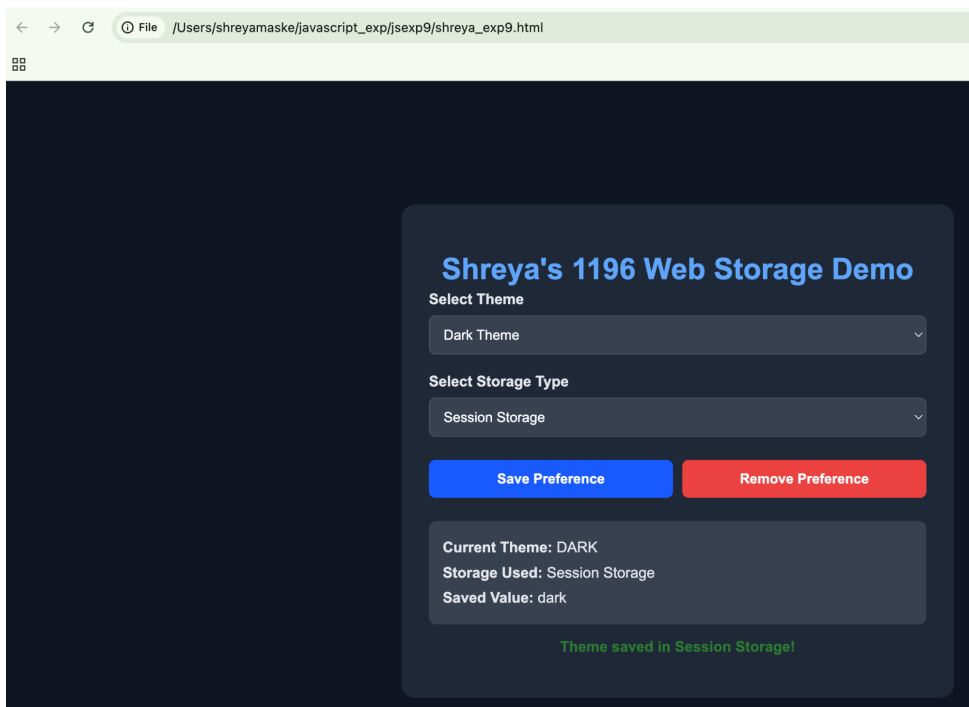
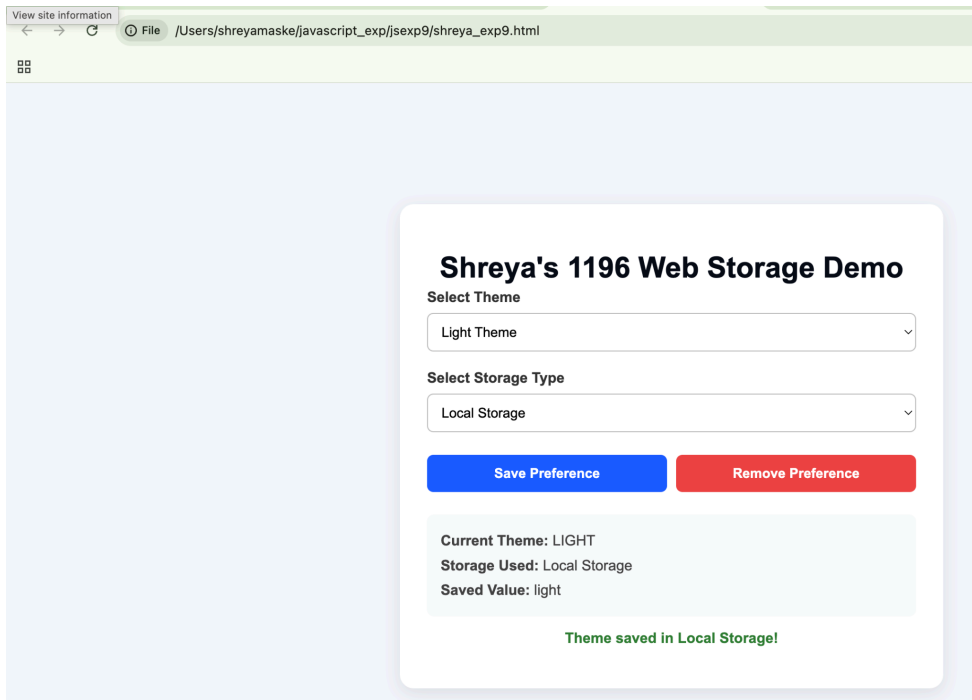
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4. Output :



5. Result/Conclusion:

Thus, local Storage and session Storage were successfully implemented to store and retrieve user theme preferences. The selected theme was applied dynamically, displayed on the webpage, automatically loaded when available, and removed successfully using JavaScript



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Case Study No.: 5

1. Case Study Title: Implementation of JavaScript Array Operations Using Built-in Methods

Case Study Program Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Array Operations</title>
  <style>
    body {
      background-color: lightblue;
    }
  </style>
</head>
<body>
<h2>JavaScript Array Operations</h2>
<label>Enter array elements:</label>
<input type="text" id="arrayInput">
<br><br>
<label>Value for Push:</label>
<input type="number" id="pushValue">
<br><br>
<label>Value for Unshift:</label>
<input type="number" id="unshiftValue">
<br><br>
<label>Splice Index:</label>
<input type="number" id="spliceIndex">
<label>Splice Value:</label>
<input type="number" id="spliceValue">
<br><br>
<label>Slice Start:</label>
```



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```
<input type="number" id="sliceStart">

<label>Slice End:</label>
<input type="number" id="sliceEnd">
<br><br>
<button onclick="arrayOperations()">Perform
Operations</button>
<p id="output"></p>
<script src="shreya_1196_case5.js"></script>
</body>
</html>
```

```
function arrayOperations() {
    let arr = document.getElementById("arrayInput").value
        .split(", ").map(Number);

    let pushValue =
Number(document.getElementById("pushValue").value);
    let unshiftValue =
Number(document.getElementById("unshiftValue").value);

    let spliceIndex =
Number(document.getElementById("spliceIndex").value);
    let spliceValue =
Number(document.getElementById("spliceValue").value);

    let sliceStart =
Number(document.getElementById("sliceStart").value);
    let sliceEnd =
Number(document.getElementById("sliceEnd").value);
    let output = "";
    output += "Original Array: " + arr + "<br><br>";
    arr.push(pushValue);
    output += "After push: " + arr + "<br>";
```



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```
// pop()
arr.pop();
output += "After pop: " + arr + "<br>";

// unshift()
arr.unshift(unshiftValue);
output += "After unshift: " + arr + "<br>";

// shift()
arr.shift();
output += "After shift: " + arr + "<br>";

// splice()
arr.splice(spliceIndex, 1, spliceValue);
output += "After splice: " + arr + "<br>";

// slice()
let newArray = arr.slice(sliceStart, sliceEnd);
output += "After slice: " + newArray + "<br><br>";

// Maximum and Minimum
let maximum = Math.max(...arr);
let minimum = Math.min(...arr);

output += "Maximum: " + maximum + "<br>";
output += "Minimum: " + minimum;

document.getElementById("output").innerHTML = output;
}
```



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3. Output:

File /Users/shreyamaske/javascript_exp/jsexp5/shreya_1196_case5.html

JavaScript Array Operations

Enter array elements:

Value for Push:

Value for Unshift:

Splice Index: Splice Value:

Slice Start: Slice End:

Original Array: 2,34,56,77,81,45,11

After push: 2,34,56,77,81,45,11,90

After pop: 2,34,56,77,81,45,11

After unshift: 1,2,34,56,77,81,45,11

After shift: 2,34,56,77,81,45,11

After splice: 2,34,56,66,81,45,11

After slice: 66,81,45

Maximum: 81

Minimum: 2

4. Result/Conclusion:

Thus, various JavaScript array operations were successfully performed and the results were displayed using user input.